

Supraja Pericherla

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SUMMARY

3+ years of experience in data science, healthcare analytics, and software development. Experienced in leading teams and collaborating with cross-functional teams and stakeholders to deliver data-driven solutions that enhance decision-making and operational efficiency. Passionate about applying machine learning and data visualization to solve real-world problems, with a focus on impactful, user-centered insights.

EDUCATION

Indiana University, Indianapolis, IN

Master's Degree in Data Science (M.S.)

Graduation: Dec 2024

BVRIT Hyderabad College of Engineering for Women

Bachelor's Degree in Information Technology (B.Tech.)

Graduation: July 2021

WORK EXPERIENCE

Indiana University, Bloomington, IN

Data Analyst

Jan 2023 - Dec 2024

- Led a team of 30 Data Science students by organizing 20 weekly meetings, while tracking and sharing the progress of 11 projects in data visualization and data analysis with stakeholders working with POLIS and FSSA (Family and Social Security Administration) clients over the course of a year.
- Developed Python Selenium Scripts to automate the download of 11,000 data rows in the form of CSV files from the Centers for Medicare & Medicaid Services (CMS) website to combine and clean the data into 200 rows to populate into POLIS's backend database.
- Followed DevOps principles to improve students' workflow across their projects, by updating weekly project tasks on GitHub and sharing their progress with stakeholders, thus improving the lab's efficiency with task completion.
- Represented the research lab for the "Innovative Product of the Year" award at the Indiana Work and Learn Conference, received an Honorable mention with 10 other teams selected out of 100+ submissions.

G Technologies, Sydney, AU

Sept 2024 - Dec 2024

Data Science Intern

- Developed a machine learning model to classify gastrointestinal images into eight distinct categories, such as polyps and ulcerative colitis, by utilizing TensorFlow and Convolutional Neural Networks (CNNs). The model incorporated advanced image processing techniques, including data augmentation and normalization, to improve the accuracy and generalizability of predictions. The dataset was processed using efficient pipelines with `image_dataset_from_directory` for seamless data loading, splitting, and preprocessing.
- Collaborated with the director to iteratively refine the CNN architecture for enhanced performance, leveraging early stopping mechanisms to prevent overfitting. Performance was evaluated through visualization of training and validation losses over epochs, along with confusion matrix analysis to identify misclassifications. The deployed model provides real-time classification capabilities for gastrointestinal endoscopic images, aiding healthcare professionals in underserved areas by improving diagnostic efficiency and accessibility.

HSBC Software Development India Pvt Ltd, Hyderabad, TS, India

Aug 2021 - Dec 2022

Software Engineer

- Led the full API development lifecycle, from requirements analysis to deployment, by engineering a robust Java-based API endpoint to manage 50K+ Australian accounts, resulting in the automation of new account registration for every 30 minutes.
- Developed and maintained APIs with Java to localize collections in MongoDB for the Indian market, in order to separate China's data, resulting in 18K+ accounts stored daily in the collections.
- Partnered with global teams to deliver data-centric projects in order to optimize international financial transactions using a Java-based API, resulting in the automation of data workflows and API development, to improve business decisions and system performance.
- Collaborated with cross-functional testing teams to provide functional test support to onboard 10K US LEAP customer accounts, by developing test cases while being their point of contact, resulting in a 40% faster response time to resolve bugs and issues testers were facing.

PROJECTS

Music Preference Analysis, Data Analytics, Indiana University

Jan 2024 - May 2024

- Examined the relationship between music genre preferences and mental health indicators like anxiety and depression by collecting and cleaning data to address inconsistencies, missing values, and standardize data formats.
- Visualized demographic trends, music preferences, and mental health scores to understand patterns, while following Exploratory Data Analysis (EDA) to examine linear relationships and use T-Tests to assess differences in mental health scores between groups with varying streaming hours.

2024 US Election Forecasting, Cloud Computing, Indiana University

May 2023 - Sept 2023

- Applied sentiment analysis to predict election outcomes by analyzing online discussions from the Reddit and Google APIs, while utilizing Amazon AWS S3 to store sentiment score calculations.
- Visualized trends with bar graphs, word clouds, and score distributions, by utilizing Exploratory Data Analysis (EDA) to uncover public sentiments about the 2024 elections.
- Evaluated model performance using machine learning techniques like cross-validation accuracy and metrics like precision, recall, and F1-score.

Hospital Care Quality in the Midwest, Scientific Clinical Data Management, Indiana University

Jan 2023 - May 2023

- Developed a predictive platform model to analyze and benchmark healthcare quality metrics across 50 hospitals in the Midwest, focusing on readmission rates, mortality rates, and patient satisfaction, predicting that mortality rates showed a discrepancy of R-squared value of 0.67 across regions.
- Utilized Excel to segment data to then use MySQL to store and query the initialized data, while implementing Python-Pandas and Scikit-Learn to handle data processing, statistical analysis, and predictive modeling.
- Created a data visualization for the Midwest, by using Tableau and Python-Matplotlib/Seaborn to generate insights based on key performance metrics across the states.
- Created an interactive web-based application in Django, enabling users to find and rank hospitals on their quality measures, thus sustaining evidence-based healthcare improvement.

CERTIFICATIONS

- [Covid19 Data Analysis Using Python](#) - Issued Jul 2020
- [Support Vector Machine Classification](#) - Issued Jul 2020
- [C with Data Structures](#) - Issued Feb 2017

SKILLS

- **Programming Languages:** Python, Java, SQL
- **Python Libraries:** Pandas, Scikit-Learn, Seaborn, Matplotlib, Selenium
- **Machine Learning & Deep Learning:** Convolutional Neural Networks (CNNs), Image processing, Image augmentation, Feature extraction, Sentiment analysis, Regression models, Predictive modeling
- **Data Visualization Tools:** Tableau, Matplotlib, Seaborn
- **Database Management:** MySQL, MongoDB
- **Cloud Computing:** Amazon AWS S3
- **Web Development:** Django, HTML, CSS
- **Project Management & Collaboration:** DevOps principles, GitHub, Cross-functional team collaboration, Stakeholder engagement